

A PROJECT TITLE
ON
Water Quality Classification Using Support Vector Machine and Random Forest
project report submitted in partial fulfilment of the requirements for the Course
MACHINE LEARNING

BACHELOR OF TECHNOLOGY
in
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by

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DEPARTMENT OF - COMPUTER SCIENCE AND ENGINEERING

Name of Title: Water Quality Classification Using Support Vector Machine and Random Forest

1. Abstract:

Water quality is an important factor for human health, agriculture, industries, and environmental sustainability. Traditional water-quality testing often requires laboratory analysis, specialized equipment, and considerable time. This project proposes a machine learning-based system to classify water samples as safe or unsafe using different water-quality parameters. The project uses a Kaggle Water Quality Dataset containing approximately 7,999 records and 21 attributes. The data is cleaned, analyzed, and preprocessed before applying Support Vector Machine (SVM) and Random Forest algorithms. The performance of both models is compared using accuracy, precision, recall, F1-score, and confusion matrix. A Streamlit-based interface is also developed to provide an easy way for users to enter water-quality parameters and obtain a classification result. The project demonstrates how machine learning can be used to analyze water-quality data and provide rapid classification for academic and experimental purposes.

2. INTRODUCTION:

Water quality plays an important role in protecting human health and maintaining environmental sustainability. Water contains various physical, chemical, and biological characteristics that determine its quality. Traditional water-quality assessment generally involves laboratory testing, which can be time-consuming and require specialized resources. Machine learning can help analyze large datasets and identify patterns in water-quality measurements.

This project focuses on developing a **binary classification system** that predicts whether a water sample is safe or unsafe. The system uses water-quality parameters as input features and applies **SVM and Random Forest** algorithms for classification. The models are trained, evaluated, and compared using different performance metrics. A Streamlit interface is also included to make the prediction system easier to use.

3. SOFTWARE REQUIREMENTS:

Requirement	Details
Operating System	Windows 10/11
Programming Language	Python 3.10
IDE	Visual Studio Code
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
ML Library	Scikit-learn
Web Framework	Streamlit
Dataset	Kaggle Water Quality Dataset
Algorithms	SVM and Random Forest
Dataset Size	Approximately 7,999 records, 21 attributes

Signature of the Guide

Course Instructor